

CANBERRA TUGGERANONG, Australia

WMO: 949250

Lat: 35.4183S

Lon: 149.0936E

Elev: 588

StdP: 94.46

Time Zone: 10.00 (AUS)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-3.7	-2.6	-5.3	2.6	1.4	-4.2	2.8	2.2	9.0	10.1	8.1	9.3	0.2	100	0.474

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.0	34.3	18.5	32.1	18.0	30.0	17.3	20.6	28.4	19.7	27.0	18.9	26.0	4.9	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	14.6	21.4	17.6	13.6	21.3	16.6	12.7	21.1	61.9	28.5	58.8	26.7	56.1	26.0	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	7.2	6.4	DB	-6.1	37.6	0.8	1.8	-6.7	38.9	-7.2	39.9	-7.6	41.0	-8.2	42.3	(4)
(5)			WB	-6.6	21.7	0.8	1.0	-7.1	22.4	-7.6	23.0	-8.0	23.6	-8.6	24.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.6	21.7	20.6	18.1	13.8	9.4	7.0	6.1	7.2	10.3	13.4	16.6	19.3	(6)
(7)		DBStd	6.09	3.38	3.11	2.97	2.84	2.60	2.41	2.36	2.27	2.63	3.09	3.48	3.18	(7)
(8)		HDD10.0	388	0	0	0	4	43	93	123	91	27	5	1	0	(8)
(9)		HDD18.3	2037	9	11	42	139	277	339	379	344	241	155	73	26	(9)
(10)		CDD10.0	1698	363	298	249	117	24	4	2	5	37	111	199	289	(10)
(11)		CDD18.3	306	114	76	33	2	0	0	0	0	0	2	22	57	(11)
(12)		CDH23.3	3584	1268	742	350	42	0	0	0	0	3	92	361	726	(12)
(13)		CDH26.7	1419	607	296	98	6	0	0	0	0	0	17	119	276	(13)
(14)	Wind	WSAvg	2.5	2.8	2.6	2.6	2.1	2.0	2.1	2.3	2.6	2.8	2.8	2.8	2.9	(14)
(15)	Precipitation	PrecAvg	702	69	70	47	41	39	65	53	58	67	60	70	65	(15)
(16)		PrecMax	1114	217	198	172	173	152	178	132	137	158	137	171	179	(16)
(17)		PrecMin	410	2	10	4	2	4	15	17	10	6	9	9	12	(17)
(18)		PrecStd	154	46	52	36	38	32	39	29	30	34	31	42	43	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.6	36.2	32.1	27.5	20.8	16.0	15.4	18.6	23.5	29.0	33.1	35.6	(19)
(20)			MCWB	19.2	20.0	17.4	15.1	12.8	10.4	10.2	10.8	13.1	14.9	17.3	17.8	(20)
(21)		2%	DB	35.2	32.6	29.3	24.2	18.8	14.4	13.8	16.2	21.2	25.9	30.0	32.2	(21)
(22)			MCWB	18.9	18.5	16.9	14.5	12.1	9.6	8.8	9.7	12.5	14.3	16.0	17.4	(22)
(23)		5%	DB	32.8	30.3	27.1	22.3	17.1	13.3	12.6	14.6	19.2	23.6	27.4	29.8	(23)
(24)			MCWB	18.6	18.0	16.4	14.0	11.3	9.2	8.1	8.9	11.5	13.6	15.7	16.9	(24)
(25)		10%	DB	30.2	27.9	25.0	20.4	15.6	12.2	11.5	13.1	17.4	21.2	24.9	27.5	(25)
(26)			MCWB	18.1	17.3	16.0	13.3	10.7	8.7	7.5	8.1	10.7	12.9	15.0	16.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.5	22.0	20.2	17.2	15.4	12.3	12.0	12.4	14.6	17.0	19.2	20.3	(27)
(28)			MCDWB	31.2	28.9	24.9	22.9	17.3	13.3	13.5	15.9	20.0	23.0	26.4	26.6	(28)
(29)		2%	WB	20.6	20.7	19.0	16.0	13.2	10.9	10.1	10.8	13.6	15.8	18.0	19.0	(29)
(30)			MCDWB	29.9	27.5	24.2	21.1	17.1	13.0	12.2	14.4	18.7	21.7	25.7	26.8	(30)
(31)		5%	WB	19.8	19.5	18.0	15.0	12.1	10.0	8.9	9.7	12.6	14.7	17.0	18.2	(31)
(32)			MCDWB	28.5	26.0	23.4	19.7	15.9	12.3	11.4	13.5	17.5	20.7	23.7	25.9	(32)
(33)		10%	WB	19.0	18.6	17.1	14.1	11.1	9.1	7.9	8.6	11.4	13.7	16.0	17.4	(33)
(34)			MCDWB	26.8	25.4	22.9	19.1	14.9	11.7	10.9	12.5	16.4	19.7	22.6	24.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	14.0	12.6	12.5	13.1	13.0	10.7	11.1	12.1	13.1	13.4	13.4	13.7	(35)
(36)			MCDBR	18.7	17.1	16.5	16.3	15.5	12.6	13.0	14.8	16.9	18.2	18.4	18.4	(36)
(37)			MCWBR	5.5	5.4	6.2	7.7	9.1	8.2	8.4	9.0	9.1	8.1	6.8	5.9	(37)
(38)		5% WB	MCDBR	15.6	13.1	12.1	12.9	13.1	10.1	10.1	12.7	14.3	14.0	14.5	14.4	(38)
(39)			MCWBR	5.3	5.0	5.4	6.6	8.1	6.9	7.0	8.3	8.1	7.0	6.1	5.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.391	0.346	0.317	0.299	0.290	0.273	0.268	0.280	0.308	0.328	0.340	0.342	(40)	
(41)		taud	2.356	2.530	2.592	2.614	2.580	2.648	2.660	2.604	2.519	2.482	2.494	2.501	(41)	
(42)		Ebn at Noon	945	967	960	924	876	871	897	931	954	975	989	997	(42)	
(43)		Edh at Noon	131	106	92	80	72	63	65	79	97	110	114	114	(43)	
(44)	All-Sky Solar Radiation	RadAvg	7.12	6.09	5.19	3.96	2.96	2.26	2.55	3.44	4.76	5.93	6.55	7.27	(44)	
(45)		RadStd	0.53	0.54	0.43	0.32	0.15	0.19	0.17	0.23	0.40	0.50	0.54	0.47	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.54	N/A	N/A	N/A	+0.89	+0.95	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)		+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+108	N/A

Nomenclature: See separate page